

GenBOM: Benchmarking and Auditing a Large Generated Product-Structure Graph

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Abstract—Can a generative model author usable engineering benchmark data? We study this question on BOMwiki-Bench, an internal benchmark whose entire content was generated by a documented model pipeline: a product-structure graph with 192,471 nodes and 263,907 edges, covering 4,879 products, 50,843 assemblies, and 136,749 parts across 53 domains. We validate it against a corpus of human-authored BOMs collected for this purpose. Three findings converge. First, at shallow depth the generated structure is indistinguishable from human structure: a matched, structure-only classifier is at chance (AUC 0.510, 95% CI [0.278, 0.734], permutation $p=0.92$), though the stratum is depth-one and small ($n=28$), so this null is weak. Second, the two populations nonetheless separate almost perfectly by text style alone, even under population matching (matched full-feature AUC 0.995; text-only AUC 1.000): observed separability is a writing-style artifact, not a structural signature. Third, completion models trained on the generated graph transfer poorly to human BOMs (fusion MRR 0.62 $\rightarrow$ 0.10 against the comparable product-held-out condition). Structural plausibility therefore does not imply distributional fidelity: generated engineering data can look right, defeat structural detectors, and still not proxy for reality. The benchmark that grounds these findings includes BOM-specific tasks (completion, where-used retrieval, domain transfer, graph repair) with leakage-resistant splits, fixed degree-matched negatives, and a feasibility-floor method for defining popularity-collapse criteria on hub-dominated graphs; the best heuristic fusion reaches MRR 0.59 to 0.63 against a popularity floor near 0.09, leaving substantial headroom. Provenance is disclosed throughout, generation artifacts are labeled rather than repaired, and a plausibility-audit framework is fixed now for future community verification on the internal BOMwiki platform. BOMwiki-Bench includes loaders, evaluators, versioned splits and negatives, the human control corpus, and the audit instrumentation.

Index Terms—bill of materials, graph benchmark, generated data, link prediction, dataset auditing

I. INTRODUCTION

Progress in graph machine learning is measured on a small set of graph families, mainly citation networks, web graphs, social graphs, and molecules [5]. Several of the most common benchmarks are small enough that a single test-node change moves reported accuracy by whole percentage points [12]. Engineered product structure is a graph family with different and underrepresented properties: directed containment rather than co-occurrence, typed hierarchy (product, assembly, part), edge attributes with operational meaning (quantities), node text, domain labels that support transfer evaluation, and reuse hubs. Generic components such as fasteners, printed circuit

boards, and connectors are shared across thousands of products, so the catalog forms one connected graph rather than a forest of trees.

We do not claim novelty for representing BOMs as graphs; that idea is established in the engineering-informatics literature [2, 17]. Our contribution is an internal, benchmark-ready release of a large product-structure graph, combined with an audit of its generated provenance. The content of the graph was authored by a generative-model pipeline (Section II). Generated data of this kind is usually either accepted uncritically or dismissed as unusable. We argue for a third position: generated engineering graphs are usable as scientific objects when they are accompanied by explicit validation instrumentation, documented failure modes, and a concrete path to human verification. Under this framing, the quality of generated data becomes an explicit target of measurement: structurally in this benchmark, and as human plausibility labels through the audit framework and community-verification layer (Section VII).

Measuring that quality against a human-authored control corpus built for this study produces the paper’s central results, and they cut both ways. At shallow depth the generated structure is statistically indistinguishable from human structure (matched structure-only AUC 0.510, at chance); yet the same populations separate perfectly by text style alone even after matching (AUC 1.000), so the detectability of generated BOMs is a writing-style artifact rather than a structural one; and completion models trained on the generated graph transfer poorly to human BOMs (fusion MRR 0.62 $\rightarrow$ 0.10 on the comparable product-held-out condition). The lesson is that structural plausibility and distributional fidelity are different properties: a generated corpus can pass structural scrutiny and still not stand in for real data. We believe this three-part result, obtained on a single artifact with its control corpus, is the most useful thing this report offers beyond the benchmark itself.

This paper contributes:

- 1) **The dataset and tooling** (Sections III and XI): canonical typed tables, a typed artifact registry, a dataset card [4], loaders and evaluators, and versioned splits with fixed negative sets.
- 2) **BOM-specific tasks with leakage-resistant evaluation** (Section IV): completion, where-used retrieval, domain transfer, and graph repair; leakage-resistant splits (product-held-out, domain-held-out, hub-stress); degree-

matched negatives; and a feasibility-floor method that makes popularity-collapse acceptance criteria well defined on hub-dominated graphs.

- 3) **Baselines** (Section V) covering structural heuristics, text, fusion, a trained logistic link predictor, and an externally validated canonical GraphSAGE reference, showing large headroom, degree leakage under naive negative sampling, and a structure-only GNN collapse on the leakage-resistant splits that identifies inductive, text-aware link prediction as the open modeling challenge.
- 4) **A decomposed generation-signature study and a transfer test** (Section VI) against a corpus of human-authored BOMs. Separating population matching from feature choice shows the two populations differ by text style but not by shallow structure, and a transfer test quantifies how poorly benchmark-trained completion carries to real BOMs.
- 5) **A plausibility-audit framework and community-verification plan** (Section VII): a frozen three-way rubric with an error taxonomy, a stratified 617-item analysis packet with full judging context, a companion artifact-review set covering all 1,350 registry artifacts, and an annotation harness, all part of the benchmark. As part of this framework we also propose a transparent, still-unvalidated BOM quality score (Section VIII).

II. PROVENANCE AND CONSTRUCTION

BOMwiki-Bench derives from BOMwiki, a public recursive BOM explorer. Two generation pipelines produced its data, and the two are disjoint.

The product graph (this benchmark). Product pages are authored as structured TypeScript files and compiled into a single graph (`ffs-graph.json`) through a graph-database round trip. The page content, including products, their multi-level decomposition into assemblies and parts, names, one-line summaries, and per-edge quantities, was generated by a documented pipeline (`scripts/generate.mjs`) using `claude-opus-4-8` (Anthropic), with the v1.0 product content committed on 2026-06-22; the model identifier and generation date are recorded in the pipeline source and repository history. Given a product-category brief, the model emits parts and assemblies with a bill of materials. Shared generic components are reused across products by construction, which produces pronounced hub structure.

A parametric parts catalog (not in the graph). A separate deterministic expander (`scripts/expand.mjs`) enumerates parametric part families with structured specifications, in the style of an industrial supplier catalog. We verified that this catalog shares no identifiers or names with the product graph. It plays no role in the benchmark beyond the provenance disclosure above.

Two properties follow. First, the topology of the graph is a large hierarchical DAG with shared substructure, but its content is model-authored; no claim of industrial ground truth is available or made. Second, generation artifacts exist. We

TABLE I
BOMWIKI-BENCH v1.0 AT A GLANCE.

Nodes	192,471
Products / Assemblies / Parts	4,879 / 50,843 / 136,749
Edges (HAS_PART, quantity-bearing)	263,907
Product domains	53
Nodes with nonempty text summaries	175,690
Artifact registry (typed)	1,350

TABLE II
TYPED ARTIFACT REGISTRY (v1.0). ARTIFACTS ARE LABELED, NOT SILENTLY REPAIRED. THE SIX CLASSES ARE DISJOINT (THE TWO QUANTITY-OUTLIER BINS PARTITION BY MAGNITUDE), SO THE COUNTS SUM WITHOUT DOUBLE-COUNTING. DUPLICATE-NAME CANDIDATES ARE NODE-LEVEL; THE OTHER FIVE CLASSES ARE EDGE-LEVEL; ALL ARE COUNTED UNIFORMLY AS ARTIFACTS.

Kind/granularity-inconsistent edges	822
Quantity outliers ($1,000 < q \leq 10,000$)	282
Duplicate-name candidates (142 nodes, 69 clusters)	142
Quantity outliers ($q > 10,000$)	93
Zero-quantity edges	7
Self-loops	4
Total	1,350

preserve and label them instead of repairing them silently (Section III-A), because they are the raw material for the audit tasks.

III. THE DATASET

Nodes carry (`id`, `kind`, `domain`, `name`, `summary`) and edges carry (`parent`, `child`, `qty`, `note`). Excluding four self-loops, the graph is a DAG. Reuse is heavy-tailed: a small population of generic parts appears in thousands of products, while most parts are specific to one product. This hub structure is realistic for engineered products, and it is also the central hazard for evaluation (Section IV). Figure 1 shows the three governing distributions. Part reuse spans four orders of magnitude, from the median part (in-degree 1) to the most-reused component (in-degree 8,026); node depth concentrates at the assembly and part levels and reaches a maximum of 5; and the 53 product domains range from 10 to 498 products. Figure 2 shows one example product decomposition.

A. The artifact registry

We scan the graph for constraint violations and preserve them under stable identifiers.

Manual inspection of the largest class shows that kind-inconsistent edges are best understood as modeling-convention violations, for example a node typed `part` that itself has children, rather than as physical impossibilities. We adopt that framing throughout. The duplicate-name clusters, in which the generator produced near-duplicates of the same conceptual part within a domain, are the product-graph analogue of the duplicate-node leakage previously documented in popular graph benchmarks [11]. These clusters are single-product-scoped: every near-duplicate group lies within one product

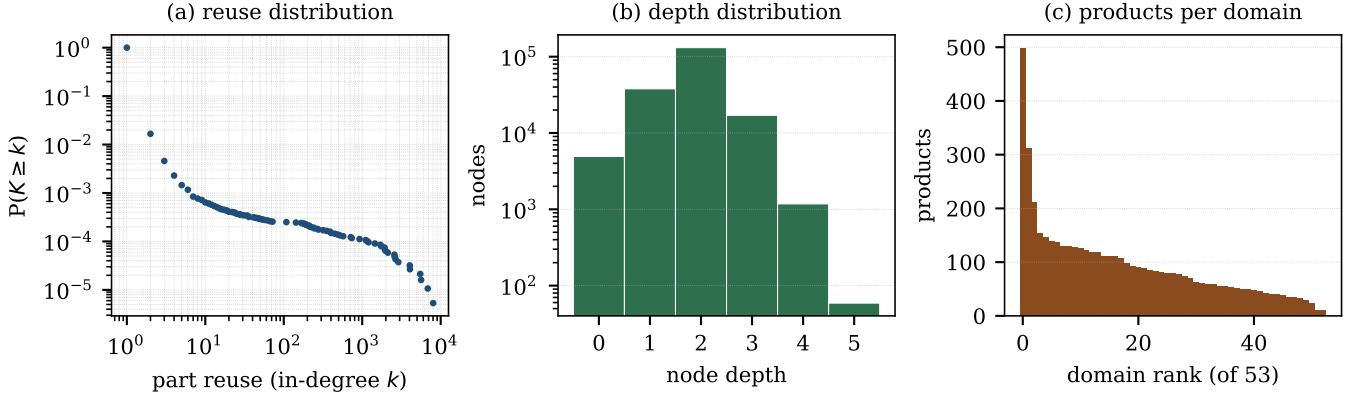


Fig. 1. Governing distributions of BOMwiki-Bench v1.0. (a) Complementary CDF of part reuse (in-degree), log-log: heavy-tailed hub structure spanning four orders of magnitude. (b) Node depth distribution (log scale): most mass at the assembly and part levels. (c) Products per domain across the 53 domains, sorted by size.

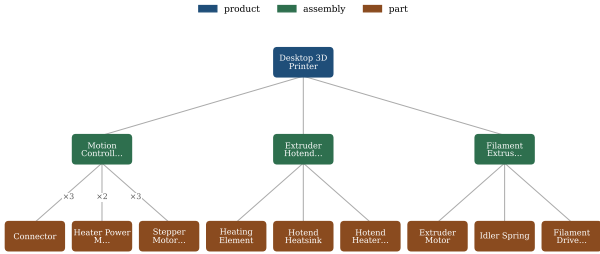


Fig. 2. An example product decomposition (*Desktop 3D Printer*), truncated to two levels and three children per node for legibility. Node color encodes kind; edge labels show quantities greater than one. Shared generic parts (connectors, bearings, fasteners) recur across many such trees, producing the reuse hubs of Figure 1(a).

subtree. Our leakage-resistant splits therefore keep each cluster intact on one side of the train/test boundary automatically. We verify this on the shipped splits: no duplicate cluster straddles train and test under the product-held-out or domain-held-out schemes, whereas the random split leaks 21 of its 28 test-touching clusters across the boundary, which is one reason random is reported only as an inflation control (Section IV).

IV. TASKS AND EVALUATION DESIGN

A. Splits

We ship versioned node- and edge-id split files: *random* (reported to demonstrate inflation, never as a headline), *product-held-out* (20% of product roots with their exclusive subtrees), *domain-held-out* (three rotations of held-out domain groups), and *hub-stress* (test edges incident to parts in the top 1% by reuse). The *hub-stress* split removes the target hub parts from training entirely: their post-split training in-degree is zero, so a model must complete edges to components it has never seen at training time. This is a deliberately adversarial regime and explains why popularity-based signals, which depend on observed training degree, collapse there (Appendix A).

B. Degree-matched negatives and the feasibility floor

BOM completion is evaluated by ranking a removed true child against 500 fixed negatives. Under uniform negatives, a pure popularity heuristic reaches MRR 0.28 overall, and 0.99 on hub edges; in that setting the evaluation measures the degree distribution rather than the models (Appendix E). Our primary protocol therefore samples negatives matched to the true child on kind and in-degree bucket. The negative sets are generated once, versioned, and shipped. We score 1,500 held-out positives per split, which keeps the 501-candidate ranking tractable across every split and negative-policy variant; deterministic baselines are computed on the fixed shipped negatives, and we attach nonparametric bootstrap intervals over the evaluation positives to quantify finite-sample stability.

Degree matching has a structural limit on hub-dominated graphs. For the rarest super-hubs (training in-degree above 500; between 287 and 389 of the 1,500 evaluation edges per standard split), too few negatives of equal or higher degree exist, so 64 to 75 edges per split are unavoidable popularity hits. We formalize this as a per-split *feasibility floor*: the popularity MRR obtained against the hardest eligible negative sets. The protocol is accepted only if observed popularity lies within 0.02 of this floor. This criterion is well defined in a setting where the naive requirement, that popularity collapse below a fixed absolute threshold, cannot be satisfied at all. On all six splits, observed popularity lands within 0.009 of its floor (Appendix A). Results are computed for the overall, non-gt500, and gt500 strata; the tables in this report show the overall stratum, and Appendix A gives the per-split hub-stratum (gt500) floor diagnostics.

C. Task definitions

Edges flagged in the v1.0 artifact registry as edge-level artifacts are excluded from scored evaluation positives across all completion splits; they remain in the training graph and in the graph-repair task, so the benchmark labels known issues without rewarding models for reproducing them.

A. BOM completion (primary; MRR and Hits@ k against matched negatives). **B. Node-kind classification** (a sanity check; near ceiling by design). **C. Domain classification and domain-held-out transfer** (53-way). **D. Where-used retrieval** (part to products). **E. Graph repair**: detect the natural artifacts of Section III-A and planted corruptions (Appendix B), each injected at a 1% rate.

V. BASELINE RESULTS

Three observations follow from Table III. First, there is substantial headroom, and it survives a trained model. The best baseline is a rank-average fusion of the popularity, text-cosine, and numeric-logistic per-candidate scores; it reaches MRR 0.59 to 0.63 on the standard splits. A logistic link predictor trained on all training positives reaches 0.44 to 0.47, well above the popularity floor near 0.09 but below the fusion and far below ceiling. We note a caveat on this comparison: the logistic is trained against random same-kind negatives but evaluated against the harder shipped degree-matched negatives, so part of its gap to the fusion is a train/evaluation negative mismatch rather than a modeling ceiling. Our GNN reference is externally validated before being trusted: the identical canonical PyG GraphSAGE stack reproduces the published ogbl-collab reference (test Hits@50 0.477 vs. 0.481) and lands ahead of the published GCN+/GIN+ band on GraphBench bluesky-quotes (MAE 0.677 vs. the 0.737 to 0.781 band) [5, 15]. (An earlier pure-Torch reference failed this gate with test Hits@50 0.004 on ogbl-collab, traced to same-kind training negatives and a one-step-per-epoch optimizer schedule; its numbers were discarded, and the audit trail is retained in the repository.) On BOMwiki-Bench, the validated model with structural features tells a split-dependent story (Table IV): on the `random` split it behaves like a degree model, reaching MRR 0.96 on `gt500` hub edges under uniform negatives and sitting at the popularity floor overall under matched negatives (0.093); on every leakage-resistant split it collapses to near-random (0.004 to 0.015 under both negative policies). The collapse has a structural cause: the held-out product and domain subtrees are *parent-side inductive*: their nodes enter evaluation with empty training neighborhoods and zero training degree, so structure-only representations degenerate, while the text-bearing logistic reaches 0.44 on the same splits. The precise open challenge for benchmark users is therefore inductive, text-aware GNN link prediction, not link prediction in general.

On hub-stress the fusion drops to 0.21, a gap of 0.41 MRR that quantifies how much completion difficulty concentrates at shared hubs. Second, the holdout scheme is a minor axis, not the dominant one. Under matched negatives the mean fusion gap between held-out and random splits is -0.020 MRR: product-held-out (-0.010) and domain-held-out-0 (-0.003) fall inside the bootstrap intervals over evaluation positives (95% half-widths up to 0.023; Appendix A), but domain-held-out-1 (-0.041) and domain-held-out-2 (-0.026) are modestly and measurably harder. Difficulty on this graph is governed primarily by the negative set and the hub structure, with a

small, split-dependent contribution from the domain holdout; users should be aware of both. Third, the results illustrate degree leakage directly, and the demonstration requires no learned model at all: the pure popularity ranker reaches MRR 0.99 on `gt500` hub edges under uniform negatives (Appendix E), drops to between 0.39 and 0.45 there under matched negatives, and overall lands within 0.009 of the feasibility floor (Appendix A). Any scorer correlated with degree inherits this inflation when the evaluation permits it; the validated GraphSAGE does exactly that on the `random` split (Table IV). This is the reason the matched protocol is the primary one. A final caveat is owed on the text signal itself: text cosine reaches MRR 0.37 to 0.39 here but only 0.066 on the human-authored corpus (Section VI), and the signature study shows generated and human naming styles are trivially separable, so the strength of text-based completion on BOMwiki-Bench is plausibly generator-specific (model-written names are stylistically consistent from parent to child in a way real SKU-style BOM text is not) rather than a property of BOM completion in general.

Other tasks. Node-kind classification is near ceiling, as expected (structural logistic regression reaches macro-F1 0.989, SAGE 0.979), and serves only as a sanity check. Domain classification from text reaches macro-F1 0.377 over 53 classes, while structure-only models remain at chance (below 0.01). Domain identity therefore lives in the text rather than the topology, which makes the domain-transfer task a text-plus-graph problem. Where-used retrieval ranks all 4,879 products as candidate parents for a query part (evaluated on 1,000 sampled parts that have at least one product ancestor). It is inherently a text or inductive task: a text baseline reaches only MRR 0.069 (Recall@10 0.123), and structural rankers (reverse-BFS, personalized PageRank) collapse to near-zero (MRR 0.003) once the query part’s own containment edges are held out. This is not a degenerate baseline but a property of the graph: 98% of parts have a single parent, so the held-out edge is the entire structural support for the answer, and where-used is meaningful for structure only on the 2% multi-parent reuse hubs. Graph repair with gradient-boosted structural features reaches ROC-AUC 0.929 and PR-AUC 0.951 on planted corruptions (26,390 instances), which leaves the natural-artifact stratum (1,350 items) as the harder open target.

VI. IS THERE A STRUCTURAL SIGNATURE OF GENERATION?

One hypothesis about a generated graph is that its structure alone reveals its origin. We test this against a corpus of human-authored BOMs collected for this study, and describe how the test was built, because its naive form gives a misleading answer.

A. The human control corpus

We harvested and normalized public, license-verified, human-authored BOMs into the benchmark schema (Appendix D). Corpus v1 contains 75 trees from machine-readable

TABLE III

BOM COMPLETION, OVERALL STRATUM, DEGREE-MATCHED NEGATIVES (MRR / HITS@1 / HITS@10), EVALUATED ON POSITIVES WITH EDGE-LEVEL REGISTRY ARTIFACTS EXCLUDED. “FLOOR” IS THE POPULARITY FEASIBILITY FLOOR (DIAGNOSTIC). “LOGISTIC” IS A TEXT+STRUCTURE LOGISTIC LINK PREDICTOR TRAINED ON ALL TRAINING POSITIVES, AVERAGED OVER THREE NEGATIVE-SAMPLING SEEDS (STANDARD DEVIATION ≤ 0.007). THE EXTERNALLY VALIDATED GRAPH SAGE REFERENCE IS REPORTED SEPARATELY IN TABLE IV.

Split	Floor	Popularity	Text cos.	Numeric LR	Fusion	Logistic
random	.094	.103/.056/.200	.387/.303/.551	.357/.236/.639	.632 /.570/.747	.435/.308/.751
product-held-out	.089	.095/.050/.195	.384/.301/.543	.356/.235/.639	.622 /.565/.731	.438/.292/.725
domain-h.o. 0	.089	.095/.048/.195	.367/.275/.551	.368/.231/.621	.629 /.566/.745	.440/.290/.736
domain-h.o. 1	.101	.108/.053/.221	.386/.307/.537	.367/.234/.625	.591 /.530/.699	.471/.329/.753
domain-h.o. 2	.085	.091/.055/.164	.376/.290/.553	.378/.257/.595	.606 /.542/.719	.458/.344/.706
hub-stress	.002	.004/.000/.000	.065/.039/.115	.063/.023/.121	.207 /.155/.286	.100/.046/.165

TABLE IV

CANONICAL GRAPH S AGE REFERENCE (STRUCTURAL FEATURES, EXTERNALLY VALIDATED ON OGBL-COLLAB AND GRAPHBENCH; THREE SEEDS, MRR STD ≤ 0.004). MRR BY SPLIT AND PROTOCOL; “UNIF. GT500” IS THE HUB SUB-STRATUM UNDER UNIFORM NEGATIVES. HUB-STRESS HAS NO TRAINING-DEGREE GT500 STRATUM BY CONSTRUCTION.

Split	matched	uniform	unif. gt500
random	.093	.271	.960
product-held-out	.006	.011	.011
domain-h.o. 0	.004	.012	.007
domain-h.o. 1	.005	.012	.007
domain-h.o. 2	.006	.015	.014
hub-stress	.005	.013	n/a

BOM files, of which 71 are flat electronics BOMs; one example is the parts list of the NASA JPL Open Source Rover. Corpus v2 contains 5 deep product assemblies (depth 3 to 6, 139 nodes in total) extracted from STEP CAD assembly structure through NEXT_ASSEMBLY_USAGE_OCCURRENCE relations, after a curation pass quarantined 48 harvested trees that turned out to be component 3D models (chip packages, connectors) rather than products.

The harvest itself produced a finding: public human-authored hardware documentation is overwhelmingly published flat. Machine-readable multi-level product BOMs are rare in public sources; deep hierarchy lives in proprietary PLM systems, in PDF assembly manuals, or in CAD assembly files, which are largely LFS-hosted and dominated by component models. As a consequence the deep human stratum is thin (9 trees), the shallow comparison is primary by design, and the 3-to-6-level trees of BOMwiki-Bench are structurally unusual among published artifacts. We state this as a matching limitation.

B. Decomposing separability: population, structure, and text

We decompose generated-versus-human separability into two effects, population matching and feature choice, because conflating them gives a misleading answer. Matching is done jointly on node-count bucket and domain (electronics), fixing depth (shallow, depth-one subtrees); on this matched $n=28$ population we vary the feature set. A unit is one root subtree. From the 71 flat human trees, 69 candidate units are extracted; matching is pooled and stratified rather than pairwise: within

TABLE V

GENERATED VS. HUMAN SEPARABILITY, DECOMPOSED (LOGISTIC; MATCHED SHALLOW POPULATION, $n=28$; THE UNMATCHED FULL-FEATURE PROBE IS A CONFOUNDED REFERENCE).

Population	Features	AUC	95% CI	perm. p
matched	structure-only	0.510	[0.278, 0.734]	0.919
matched	full (with text)	0.995	[0.969, 1.000]	$< 10^{-4}$
matched	text-style only	1.000	[1.000, 1.000]	$< 10^{-4}$
unmatched	full (with text)	1.000	n/a	n/a

each supported (domain, node-count-bucket) stratum, an equal-count sample with a fixed seed draws the same number of human and BOMwiki-Bench units (nine per source in the electronics 5-to-10-node bucket and five per source in the 11-to-25 bucket), giving $14+14 = 28$. Human candidates in the 26-to-60-node bucket are unused because BOMwiki-Bench offers no shallow support there; the deep stratum ($n=16$) applies the same construction to the deep corpora. Structure-only features are quantity-distribution moments, branching statistics, duplicate-child rate, and quantity roundness; the full set adds text-style features (name and summary statistics). Grouped cross-validation keeps all units from one source repository (human) or one product root (BOMwiki-Bench) in the same fold, and label permutation is restricted within the matched strata (an exact space of 1.2×10^7 relabelings).

The decomposition overturns a natural but wrong reading. On the matched population a structure-only classifier is at chance (AUC 0.510, within-stratum $p=0.919$; GBDT agrees at 0.441), yet the same matched population with text features is separated almost perfectly (AUC 0.995, $p < 10^{-4}$), and text-style features alone reach AUC 1.000. A matched, full-feature classifier is thus nearly as separable as the raw unmatched probe (1.000): population matching removes almost none of the separability. What removes it is dropping text. Generated and human BOMs are therefore trivially distinguishable by writing style but not by shallow structure: the separability is a text-style artifact, not a structural generation signature. The methodological lesson is accordingly about feature ablation, not population matching: a structural-detectability claim must restrict to structural features, or it measures writing style rather than generated structure.

Three qualifications bound the structure-only null. A

within-stratum label-permutation test (100,000 relabelings; Appendix F) confirms it, $p=0.919$ (logistic) and $p=0.561$ (GBDT). With $n=28$ and a matched-AUC interval of width ≈ 0.46 , the test is powered only for large effects (AUC $\gtrsim 0.75$). And the stratum is restricted to depth-one subtrees, whose branching and quantity moments carry little structure, so a shallow structural null is close to guaranteed by construction; the deep stratum ($n=16$) is uninformative and has too few exchangeable units for a valid permutation (Appendix G).

C. Transfer to real BOMs

The complement of the signature question is whether completion skill learned on BOMwiki-Bench carries to real product structure. We train the completion baselines on the full BOMwiki-Bench graph and evaluate them on held-out edges of the human corpus, with degree-matched negatives built in the human-candidate space. As the in-distribution anchor we use the `product-held-out` split, the condition most analogous to encountering a novel product. Transfer is poor: the fusion ranker falls from MRR 0.622 in-distribution to 0.095 on the human corpus (flat 0.099, deep 0.041), and the trained logistic from 0.438 to 0.033. Only text-cosine retains any signal, and only on the deep trees (0.264). Two protocol caveats bound the comparison. The human candidate space is smaller than the benchmark’s: flat-corpus positives are ranked against 496 negatives on average (close to the benchmark’s 500, with a uniform fallback on 9% of edges), but deep-corpus positives average only 117 negatives with fallback on 94% of edges. Smaller pools inflate human-side MRR, so the transfer gap is, if anything, understated; the deep-corpus numbers additionally rest on five trees and should be read as indicative only. Skill on BOMwiki-Bench does not, at present, transfer to human BOMs. This is consistent with the signature result: generated and real product structure form distinct distributions, separable by style, and competence on one does not carry to the other. It also bounds the benchmark’s external validity, which we state plainly in Section X.

VII. A PLAUSIBILITY-AUDIT FRAMEWORK AND COMMUNITY VERIFICATION

Because the content is generated, plausibility is an empirical question rather than an assumption. This benchmark does not yet include human plausibility labels. It does include the complete instrumentation needed to produce them, and we describe the deployment plan because it is part of the benchmark’s design.

A. Audit framework

The framework comprises three artifacts (full detail in Appendix C). First, a frozen three-way rubric (`plausible`, `implausible`, `cannot-judge`) with a six-way error taxonomy and calibration instructions with worked examples. Second, a stratified audit packet of 617 analysis items (sampled edges and nodes plus 201 registry artifacts) and 20 separate calibration items, together with a companion artifact-review set that covers all 1,350 registry artifacts; every item carries

the full judging context (parents, children, quantities, sibling components, and product context). Third, a self-contained annotation harness that presents items, records labels and timing, and enforces two integrity rules: no label is ever suggested or prefilled, and items are displayed verbatim. An analysis pipeline (inter-annotator agreement with a pre-registered gate of Cohen’s $\kappa \geq 0.6$, stratified Wilson confidence intervals, error-taxonomy summaries, and structural correlations) is included.

B. Deployment: continuous verification on the BOMwiki platform

The graph audited here is also served as a live website (`bomwiki.com`), which users already browse product by product. Rather than a one-time expert annotation, we plan to execute the audit as a continuous community-verification layer embedded in the site: each displayed BOM edge carries a lightweight prompt (“does this part belong in this assembly?”) mapped to the rubric above. Quality is obtained statistically rather than by credential screening [13]: multiple independent judgments per item, gold-standard items with known answers to calibrate and weight contributors, and supermajority thresholds before an item is marked verified. Verified status will be surfaced on the site itself, and accumulated labels will be added as versioned additions to the benchmark (v1.1 and later), with the v1.0 artifact of this report remaining frozen. Effort will be prioritized where labels are most informative: the artifact registry, high-traffic products, and high-reuse hub components. Two deployment requirements follow from the site being a drifting live surface rather than the frozen snapshot. Labels must be keyed to stable artifact identifiers so that v1.1 labels attach to the correct v1.0 edge even as the live content changes; and the site must disclose to users that content is model-generated and unvalidated, so the “not for real sourcing or engineering” caveat of Section X is a property of the surface itself, not only of this report.

Until such labels exist, the plausibility of the generated content is unvalidated, and we state this plainly in Section X. The framework is fixed now so that the labeling protocol, strata, and integrity rules are frozen before any labels are collected, in the same pre-registered spirit as the rest of the benchmark.

VIII. A TRANSPARENT BOM QUALITY SCORE

We ship `quality_score(subtree)`, a rule-based score with pre-registered equal weights over the artifact classes of Section III-A: cycles and self-loops, quantity outliers, zero quantities, kind/granularity inconsistencies, duplicate-name children, and missing text. As an example, the *Smart-phone* product tree (121 nodes) scores 0.50 and is flagged for quantity outliers, granularity inconsistency, and missing text. The score is proposed, not validated: its calibration against human plausibility judgments is designed to come from the community-verification labels of Section VII, and the weights will remain frozen until that validation exists.

IX. RELATED WORK

Graph ML benchmarks. OGB standardized realistic datasets, splits, and evaluators [5]; later work argues for text-attributed and controlled graph families [8, 10]. The most recent comprehensive suite, GraphBench [15], standardizes protocols and out-of-distribution splits across domains that include chip design and electronic circuits; its engineered-artifact graphs are component netlists, however, and it contains no product-structure or containment-hierarchy family. None of these covers physical product decomposition. **BOM and product-structure graphs.** Network analyses of BOMs [2], BOM database benchmarking [17], and BOM comparison methods [16] establish that BOMs-as-graphs is prior art; none pairs a large product-structure graph with graph-learning tasks and a hardened evaluation protocol of the kind we build here. **Supply-chain GNNs.** SupplyGraph and its successors model firms, transactions, and planning [1, 18], the complementary graph family to internal product decomposition. **SBOM and HBOM.** Large public software-BOM corpora [14] and BOM standards [3] demonstrate demand for machine-readable component inventories; BOMwiki-Bench is a physical-product, graph-learning counterpart. **Validation of LLM-generated knowledge graphs.** Work on LLM KG construction and graph judging addresses generic triple quality [6, 19]; BOMwiki-Bench contributes domain-structural validation and, to our knowledge, the first matched-control test for structural generation signatures in engineering graphs. **Crowd-sourced data validation.** Redundancy-based aggregation of non-expert judgments and gold-question quality control are established practices [13]; our contribution is their application to generated engineering-graph verification with a pre-registered rubric and frozen strata. **Benchmark-validity critique.** Studies of duplicate-node leakage and evaluation pitfalls in GNN benchmarks [9, 11, 12] motivate both our duplicate audit and our negative-sampling protocol. For link prediction specifically, HeaRT [7] shows that the choice of negative set governs reported rankings and argues for harder, structure-aware negatives; our degree-matched negatives and feasibility floor are a product-graph instance of that argument, targeted at the popularity-collapse failure mode on hub-dominated graphs. Dataset documentation follows Datasheets practice [4].

X. LIMITATIONS AND INTENDED USE

(1) **Provenance.** The content is model-generated. BOMwiki-Bench is a benchmark object, not industrial ground truth, and must not be used as real sourcing or engineering data. (2) **Plausibility is not yet human-validated.** This benchmark includes the audit framework and instrumentation, not labels; until community verification produces labels, statements about content quality rest on structural constraints and the artifact registry alone, and the quality score is proposed rather than validated. (3) **Statistical power and design of the signature study.** The matched shallow test has $n=28$ (powered only for large effects) and is restricted to depth-one subtrees that carry little structure; the deep stratum is thin (9 human trees) because public deep human BOMs are

rare. The feature decomposition (Section VI) already shows that text, not population matching, drives separability; making the deep stratum interpretable requires a corpus expansion that remains future work. (4) **Single generator.** A single model (claude-opus-4-8) authored the content; the pipeline is documented and retained internally, which enables controlled regeneration. (5) **Artifacts are part of the benchmark.** The registry is versioned; repairs would constitute a new dataset version. (6) **Community labels introduce their own biases.** Visitor judgments will over-represent popular products and domains; released label versions will report per-stratum coverage. (7) **Transfer to real BOMs is poor.** The transfer test (Section VI) shows completion skill learned on BOMwiki-Bench does not carry to the human corpus (fusion MRR $0.62 \rightarrow 0.10$ against the product-held-out anchor). Results on BOMwiki-Bench should therefore be read as competence on generated product structure, not as a proxy for real BOMs; the human sample we can collect is also small, so this bound is itself provisional. (8) **The GNN reference is structure-only.** Our validated GraphSAGE baseline uses structural features without text, and it collapses on the leakage-resistant splits because they are parent-side inductive (Section V). Whether a text-featured, inductive GNN closes the gap to the fusion is the benchmark’s open modeling question; until one is reported, GNN capability on BOMwiki-Bench is bounded only from below.

XI. CONCLUSION

BOMwiki-Bench contributes a large, typed, quantity- and text-bearing product-structure graph under full provenance disclosure; an evaluation design hardened against the graph’s own hazards (hubs, duplicates, degree leakage), including a feasibility-floor acceptance method; baselines with large headroom; a matched-control null result on structural generation signatures; and a pre-registered audit framework whose execution is planned as continuous community verification on the BOMwiki platform that hosts the graph. The general position we argue for is that generated engineering data becomes usable as a scientific object when its provenance, artifacts, failure modes, and verification path are treated as first-class parts of the benchmark.

Data availability

BOMwiki-Bench is proprietary to ERP•AI and is not released under an open license. It is maintained internally on the BOMwiki platform (<https://bomwiki.com>), which hosts the dataset, splits, fixed negatives, the human control corpus with per-tree source licenses, the audit packet and annotation harness, and the loaders and evaluators. All results in this report are computed on the frozen v1.0 snapshot, source commit 418b44d3, data/v1.0 SHA-256 c01f47e9...76794, frozen 2026-06-22; the live site continues to change and is not the benchmark. The frozen snapshot is retained internally for reproducibility; no external archival identifier is assigned.

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APPENDIX A

FEASIBILITY-FLOOR CONSTRUCTION AND PER-SPLIT DIAGNOSTICS

For each positive evaluation edge ($p \rightarrow c$), the eligible negative pool consists of nodes of the same kind as c , excluding c itself, the existing children of p , and p . The feasibility floor is computed by ranking c by training-degree against the 500 highest-train-degree eligible negatives, the adversarially hardest set a degree-matching sampler could produce. If c is a super-hub with fewer than 500 eligible negatives of equal or higher degree, a popularity ranker unavoidably places c at or near rank one; the floor aggregates these unavoidable hits. Acceptance requires the observed popularity MRR under the shipped matched negatives to lie within 0.02 of the floor. The 0.02 tolerance is a fixed absolute margin set before the observed MRRs were computed. It is roughly one fifth of the floor value on the standard splits (near 0.10) and small relative to the gap between the shipped matched negatives and the uniform alternative (popularity MRR near 0.10 versus 0.28 overall, a 0.18 difference; Appendix E). It therefore accepts a negative set that reaches the adversarial floor without admitting a materially easier one.

TABLE VI
PER-SPLIT FLOOR DIAGNOSTICS (1,500 EVALUATION EDGES PER SPLIT). “>500 EDGES” COUNTS POSITIVES WHOSE TRUE CHILD HAS TRAINING IN-DEGREE ABOVE 500; “UNAV. H@1” COUNTS UNAVOIDABLE POPULARITY HITS AT RANK 1.

Split	>500 edges	unav. H@1	floor MRR	observed pop. MRR	margin	accept
random	363	71	0.0943	0.1026	+0.0083	yes
product-held-out	344	67	0.0885	0.0947	+0.0062	yes
domain-held-out 0	344	64	0.0887	0.0952	+0.0065	yes
domain-held-out 1	389	72	0.1009	0.1077	+0.0068	yes
domain-held-out 2	287	75	0.0852	0.0914	+0.0062	yes
hub-stress	0	0	0.0020	0.0040	+0.0020	yes

All margins are below the 0.02 acceptance threshold. Popularity computations use training-split degrees only; full-graph degrees would leak held-out edges into the ranker. The “>500 edges” column counts *training-split* in-degree, which is why *hub-stress* reads 0 even though its test children are among the highest-reuse parts in the full graph: the split removes those parts from training entirely, so their training in-degree is zero and no positive qualifies as a training-degree super-hub. The same mechanism drives the near-zero *hub-stress* floor, since a popularity ranker has no training signal for an unseen child. Within the *gt500* stratum itself, popularity under the shipped matched negatives reaches MRR 0.390 to 0.451 per standard split; these are the unavoidable-hit contributions that the floor construction accounts for.

APPENDIX B

CORRUPTION-INJECTION PROTOCOL (GRAPH REPAIR)

On a working copy of the graph, each corruption type is injected at a 1% edge rate with a fixed seed, and every injected element is labeled: (1) *reversed containment*: an existing edge is duplicated in the reverse direction, creating a cycle; (2) *self-loops*: a node is attached to itself; (3) *quantity outliers*: an edge quantity is multiplied by 1,000; (4) *kind-violating edges*: a containment edge is added whose endpoint kinds violate the product/assembly/part convention; (5) *cross-domain child swaps*: a child is replaced by a random same-kind node from a different domain. The detection task is binary (injected vs. original) over the union, evaluated with ROC-AUC and PR-AUC. The reported baseline (histogram gradient boosting over structural features) reaches ROC-AUC 0.929 and PR-AUC 0.951 on 26,390 instances (7,917 test). Natural artifacts from the registry are evaluated as a separate, harder stratum and are not mixed into the planted-corruption metrics.

APPENDIX C

AUDIT RUBRIC, CALIBRATION ITEMS, AND INTEGRITY RULES

Rubric (three-way, forced choice). *plausible*: an engineer reviewing this BOM line would accept it without a second look; generic shared parts under almost any product are plausible. *implausible*: clearly wrong (physically impossible containment, wrong-domain part, absurd quantity in context, self-referential, or nonsensical part). *cannot-judge*: requires expertise or context the item does not provide; a legitimate answer that must never be forced into a guess.

Error taxonomy (primary tag, only when implausible). *impossible-containment*, *wrong-domain-part*, *implausible-quantity*, *redundant-duplicate-child*, *vague-nonphysical-part*, other (free text).

Judgment rules fixed in advance. Quantities are judged against product context (260 surface-mount passives on a main-board is plausible; 260 wheels is not). The item is judged, not the whole tree. Kind/granularity inconsistencies are convention violations and are tagged other unless something is also semantically wrong.

Procedure and integrity rules. Annotators label the 20 calibration items independently, then discuss rubric interpretation only; afterwards the rubric is frozen and independent labeling begins (sessions capped at two hours). The harness never suggests or prefills a label. Agreement is gated at Cohen’s $\kappa \geq 0.6$; if the gate fails, the rubric wording is revised and re-calibrated on a fresh subsample, and agreement is never manufactured by discussing individual answers. Adjudicated labels are stored separately; raw labels are released unmodified. In the community deployment, the same rubric is presented as a lightweight per-edge prompt, with redundancy, gold items, and supermajority thresholds substituting for annotator training.

APPENDIX D

HUMAN CORPUS SOURCE TABLE AND LICENSES

Corpus v1 (75 trees; 1,793 nodes). Machine-readable BOM files from public repositories, license-verified before inclusion; 71 trees are flat (depth-1) electronics BOMs and 4 are multi-level. License distribution: MIT (33), GPL-3.0 (13), Apache-2.0 (7, including the NASA JPL Open Source Rover parts lists), CC-BY-SA-4.0 (6), BSD-2-Clause (3), CC0-1.0 (3), CERN-OHL-S-2.0 (3), CERN-OHL-P-2.0 (2), CERN-OHL-W-2.0 (2), CC-BY-4.0 (1), BSD-3-Clause (1), CC-BY-SA-3.0 (1).

A curation pass quarantined 48 additional harvested STEP trees that were component 3D models (chip packages, connectors, LED bodies) rather than product assemblies; the quarantine and its decision rules ship with the corpus as an audit trail. Per-tree *provenance.json* files record source URL, license, retrieval date, and normalization notes. No model-composed text appears in any corpus data field; normalization was deterministic parsing only.

TABLE VII
BOM COMPLETION, OVERALL STRATUM, UNIFORM NEGATIVES (MRR / HITS@1 / HITS@10). SAGE IS THE EXTERNALLY VALIDATED CANONICAL REFERENCE (TABLE IV).

Split	Popularity	Text cos.	Numeric LR	Fusion	SAGE (PyG)
random	.282/.271/.293	.389/.305/.555	.374/.324/.451	.093/.073/.109	.271/.260/.280
product-held-out	.263/.257/.265	.373/.284/.539	.356/.309/.427	.065/.050/.075	.011/.001/.003
domain-h.o. 0	.258/.255/.258	.385/.298/.565	.352/.305/.426	.069/.051/.085	.012/.001/.001
domain-h.o. 1	.296/.291/.299	.381/.297/.542	.395/.355/.467	.067/.052/.079	.012/.001/.002
domain-h.o. 2	.244/.237/.248	.382/.289/.560	.345/.293/.425	.065/.048/.080	.015/.001/.005
hub-stress	.002/.000/.000	.061/.035/.113	.013/.005/.019	.003/.000/.000	.013/.000/.000

TABLE VIII
CORPUS V2 (5 TREES; 139 NODES), EXTRACTED FROM STEP ASSEMBLY STRUCTURE.

Tree	Source project	License	D	N
Rover motor ctrl. asm.	nasa-jpl/open-source-rover	Apache-2.0	3	63
Toolhead Bowden asm.	VoronDesign/Voron-0	GPL-3.0	4	31
Accelerometer mount	VoronDesign/Voron-0	GPL-3.0	6	22
Enclosure asm.	iley/kamaji	MIT	3	12
Idler asm. (Voron V0)	MasturMynd/Pandora	GPL-3.0	3	11

APPENDIX E UNIFORM-NEGATIVE RESULTS (INFLATED; FOR CONTRAST)

The same baselines as Table III, evaluated against 500 uniform random negatives. Popularity inflates roughly three-fold overall, and reaches MRR 0.99 on the `gt500` hub sub-stratum, which is the degree-leakage effect the matched protocol removes. The SAGE column reports the externally validated canonical reference of Table IV under uniform negatives: it tracks popularity on the `random` split (0.271 overall, 0.960 on `gt500`) and collapses on the leakage-resistant splits for the parent-side-inductive reason given in Section V. (An earlier pure-Torch reference that failed the external sanity gate was removed from this table; its numbers remain in the repository audit trail.) Rank-average fusion degrades sharply here (MRR 0.065 to 0.093) even though each of its three components scores well individually (0.28 popularity, 0.39 text, 0.37 numeric-logistic). This is a property of rank fusion, not a weak component: under uniform negatives the three component rankings become discordant, each placing the true child at the top of a different candidate set, so averaging their ranks dilutes the rank-one hits that each component achieves alone. Under matched negatives the components are concordant (all struggle on the same hard candidates) and the fusion is instead the best baseline. The inversion between the two protocols is itself evidence that the choice of negative sampling, not the model, dominates naive evaluations on this graph.

APPENDIX F SIGNATURE PERMUTATION-NULL DISTRIBUTIONS

For the matched $n=28$ shallow stratum we recompute AUC from fixed out-of-fold scores under 100,000 label permutations

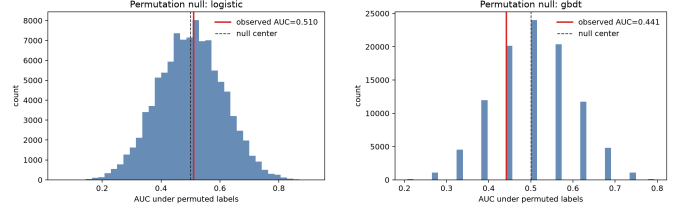


Fig. 3. Permutation-null AUC distributions over 100,000 label permutations of the matched $n=28$ units; red line marks the observed AUC. Both observed values fall inside the bulk of their null, consistent with no structure-only signature.

restricted within the matched strata (an unrestricted permutation over matched data is not exchangeable). Figure 3 shows the resulting null distributions; the observed structure-only AUCs (logistic 0.510, GBDT 0.441) sit near the center of their nulls, giving two-sided tail probabilities 0.919 and 0.561.

APPENDIX G DEEP-STRATUM SIGNATURE METRICS (UNINFORMATIVE; FOR COMPLETENESS)

The deep matched comparison (BOMwiki-Bench deep subtrees vs. the 9 deep human trees, matched on node-count bucket, depth bucket, and domain where supported; $n=16$) produced: logistic AUC 1.000 (95% CI [1.000, 1.000]) and GBDT AUC 0.344 (95% CI [0.111, 0.604]) under grouped 5-fold cross-validation. We flag two readings rather than assert one. The logistic score is out-of-fold, which classic overfitting degrades rather than inflates, so a perfect out-of-fold separation on a stratum where BOMwiki-Bench trees are elsewhere described as structurally unusual could be a genuine signature; on the other hand, the two classifiers contradict each other sharply on the same 16 samples (GBDT 0.344), which is equally consistent with model-class failure or instability at $n=16$. We do not have a permutation test at this depth (the deep matched design has too few exchangeable units for a stable null); computing one on a larger deep human corpus is the way to resolve the ambiguity, and we defer interpretation until then (Section X). We publish the raw numbers so that neither reading can later be quoted selectively.